



BAHRIA UNIVERSITY KARACHI CAMPUS

DIGITAL CAR PARKING COUNTER SYSTEM

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Subject: Digital Logic Design

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ABSTRACT

This project presents the design and implementation of a digital car parking counter system using basic digital logic components. IR sensors are used to detect vehicle entry and exit, while JK flip-flops perform counting operations. The counted value is decoded and displayed on a common cathode seven-segment display using a BCD-to-seven-segment decoder. The project demonstrates the practical application of sequential and combinational logic.

CHAPTER 1: INTRODUCTION

Background

Car parking management systems are widely used to monitor vehicle flow in parking areas. Digital counters play a vital role in such systems by tracking the number of vehicles entering and leaving a parking lot. This project focuses on implementing a basic car parking counter using discrete digital logic ICs.

Literature Review

Counters implemented using JK flip-flops are well documented in digital electronics textbooks. JK flip-flops provide toggling functionality, making them suitable for synchronous and asynchronous counter designs. According to standard digital design principles, combinational logic is used alongside flip-flops to generate valid counting sequences. The use of BCD-to-seven-segment decoders such as the 7448 IC simplifies numeric display interfacing [1][2].

Problem Statement

Manual vehicle counting is inefficient and prone to error. An automated digital system is required to accurately count vehicles entering and exiting a parking area.

Aims and Objectives

1. Design a car parking counter using JK flip-flops.
2. Detect vehicle movement using IR sensors.
3. Display the count on a seven-segment display.
4. Understand sequential logic implementation.

Scope of Project

The project is limited to single-digit counting (0–9). The design can be extended to multidigit counters by cascading additional flip-flops and display units.

Environmental Aspects of Project

The project uses low-power ICs and operates on a 9V battery, making it energy efficient. No hazardous materials are involved, and the system produces negligible electronic waste.

CHAPTER 2: DESIGN AND METHODOLOGY

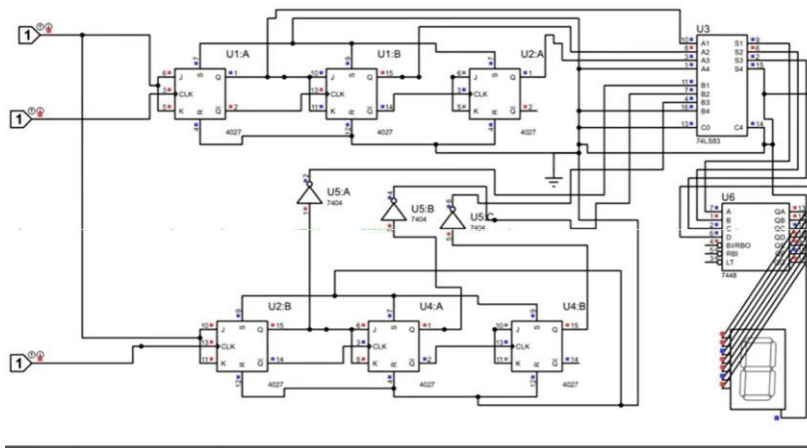
Components Used

- Breadboard (2)
- IC 7404 NOT Gate (3)
- IC 7448 BCD to Seven Segment Decoder (3)
- IC 4027 Dual JK Flip-Flop (3)
- IR Sensors (2)
- Capacitors 10 μ F (4)
- 9V Battery with Connector (1)
- Seven Segment Display (Common Cathode) (1)

Circuit Diagram

The circuit consists of:

- JK flip-flops configured as a counter
- NOT gates for logic conditioning
- 7448 IC for BCD to seven-segment decoding
- Seven-segment display for visual output



Working Principle

When a vehicle passes in front of the IR sensor, a pulse is generated which acts as a clock input to the JK flip-flops. The flip-flops toggle accordingly and generate a binary count. This binary output is decoded by the 7448 IC and displayed on the seven-segment display.

Implementation

1. Power supply connections were established using a 9V battery.
2. IR sensors were connected to generate clock pulses.

3. JK flip-flops (4027) were configured in toggle mode.
4. NOT gates (7404) were used for signal inversion where required.
5. Flip-flop outputs were connected to the 7448 decoder.
6. The seven-segment display was interfaced with the decoder.
7. Capacitors were used for noise filtering and signal stability.
8. The complete circuit was tested and verified.

Results

- The system successfully counted objects detected by the IR sensor.
- The seven-segment display correctly showed numbers from 0 to 9. □
The counter incremented reliably with each detection event.

CHAPTER 3: CONCLUSION

The digital car parking counter system was successfully implemented using basic digital logic components. The project effectively demonstrates the use of JK flip-flops, logic gates, and display decoders in a real-world application.

Recommendations

- Extend the system for multi-digit counting.
- Replace IR sensors with laser sensors for improved accuracy.
- Implement PCB design for better reliability

REFERENCES

- [1] M. Morris Mano, *Digital Design*, Pearson Education
- [2] Thomas L. Floyd, *Digital Fundamentals*, Pearson
- [3] YouTube Video Reference: Digital Counter Using JK Flip-Flops

Digital Logic Design Lab Assessment Rubric For Project Evaluation

Level: Mechanism (P4)					
Criteria	Achievement Level				Total
	Excellent (5)	Good (2.5)	Average (1.25)	Poor (0)	
Approach towards task completion	Demonstrated an excellent and systematic approach towards the task completion.	Demonstrated a good approach towards the task completion but partially unsystematic.	Demonstrated an ambiguous and unsystematic approach towards the task completion.	Didn't not show any rational approach to carry out the assigned task.	
Skill of Handling Tool (Software)	Outstanding display of confidence and skill in handling of tools. Follow procedures precisely and flawlessly.	Proficient in handling of tools but require focus & concentration. Follow procedures effectively.	Less proficient in handling tools. Even with focus commits mistakes. Require corrections while following procedure.	Unable to handle the tools. Can't follow the procedure.	
Proficiency in Calculating Experiment Parameters and Obtained Results	Calculate the parameters without any mistake. Obtained results are accurate.	Calculate parameters with adequate confidence. Obtained results possess hardly any mistake	Calculate experiment parameters with mere confidence and results are also inconsistent.	Couldn't calculate parameters. Results are missing.	
Discussion on results and conclusion	Thoroughly discussed the recorded observations and drew correct conclusions based on theoretical concepts.	Appropriately discussed the recorded observations but drew partially correct conclusions based on theoretical concepts.	Discussed the recorded observations partially and drew incorrect conclusions i.e., conflicting the theoretical concepts.	Discussion on the recorded observations and conclusions are either completely wrong or missing.	

Marks Obtained: _____/20

Teacher's signature: _____